

A glimpse of what's in store

As far as flat screens go, the future promises to be interesting. Here is a sample of how current research is likely to revolutionise the future.

- The April issue of the journal *Nature* features research by a Rice University team that could lead to unimaginably thin flat screens. They have discovered a process that enables the splitting of carbon nanotubes to make nano-ribbons at room temperature. These ribbons are basically conductive films, so they can be used to make flexible electronic displays.
- The Oregon State University, in association with HP, holds patents for a transparent zinc-tin oxide semiconductor, which is environmentally benign, performs well and can be made from inexpensive, common materials. These semiconductors can be used to make stable see-through electronic components. This would mean very thin and cool displays that will lend themselves to many futuristic applications.
- Another team at the University of Illinois has created a new process to create ultra-thin inorganic LEDs that shine brighter and last longer than conventional LEDs. "Our goal is to marry some of the advantages of inorganic LED technology with the scalability, ease of processing and resolution of organic LEDs," said the professor who leads the team. "By printing large arrays of ultra-thin, ultra-small inorganic LEDs and interconnecting them using thin-film processing, we can create general lighting and high-resolution display systems that otherwise could not be built with the conventional ways that inorganic LEDs are made, manipulated and assembled." If and when this is commercialised, you may be able to wrap your TV and carry it in your handbag!

well as related peripherals because unlike older technologies, these combine both audio and video capabilities. Also Blue-ray signals are compatible only with HDMI. Hence almost every manufacturer includes HDMI input plugs in their flat screen displays. Some go a step ahead and provide extra HDMI slots such as side-mounted ones for connecting camcorders, etc.

Eco-friendly and energy-efficient displays

Energy efficiency and eco-friendliness is one key area of innovation for most flat screen manufacturers as it has become a critical issue these days. For example, this September the California Energy Commission released the country's first rules mandating energy efficiency for TVs. TV sets sold in California will be required to reduce electricity consumption by 30 per cent by 2011 and 50 per cent by 2013. This came as a shock to many flat screen manufacturers, especially since plasma TVs are known to be major energy hogs! Definitely other states and countries will follow the example set by California.

This will provide an impetus for flat screen manufacturers to increase

their efforts towards energy efficiency and eco-friendliness. The key players are already committed to this cause and making important moves by shifting to LEDs, OLEDs, etc.

Panasonic's Neo PDP display technology is a good example of efforts towards energy efficiency. The company claims it reduces power consumption by 50 per cent compared to earlier and comparable models. The company exhibited products using this technology at the Consumer Electronics Show (CES) 2009, touting its energy consumption to be at par with or lower than LCD displays. The new plasma screen technology has larger cells to emit more light at lower power. That makes the screen 50 per cent brighter than normal, using the same amount of power. Neo PDP also has a feature called 'infinite black,' wherein parts of the screen shut off when displaying dark scenes, thereby conserving energy.

Easy viewing

Consumers are also beginning to pay a lot of attention to ergonomics and health aspects. They look for convenient wall-mounting options supporting good viewing angles. Similarly, they look for features that make the screen-

viewing experience less harmful to their eyes. Flat screen display makers are therefore including feature like anti-reflective filters to be easy on the viewer's eyes.

Integration is important

Integrated Digital Television (IDTV) is now catching on in Asia too. These are TVs that have tuners built into them. Usually the tuner includes a digital TV tuner and MPEG-2/MPEG-4 video decoder to receive high-quality 1080i HD broadcasts. Samsung, Philips, Sony and LG have IDTV models on sale.

SED is better

The resurgence of a project that was abandoned some years ago due to legal hassles promises better display technology for tomorrow's flat screens. Canon and Toshiba have announced that they are going to renew their work on surface-conduction electron-emitter display (SED) technology. SED extends the principles of CRT displays (minus the disadvantages) to make high-definition flat screens. SED technology has superior picture quality with wide-viewing angles and better colours and blacks. SED TVs have a response time of less than 1 ms and an actual contrast ratio of 50000:1.

Odds and ends

While the above captures the main technological waves in the flat screen market, individual manufacturers are also throwing in their share of miscellaneous features such as the ability to read SD cards directly, in-built web cameras, 3D views, etc. After making a shortlist of brands based on the main parameters such as size, contrast ratio, response time, quality of speakers, HDMI and low power consumption, dig deeper to find out what tiny, nice-to-haves the manufacturers are offering you. ●

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